Starch test experiment

GRADE 9 | INTEGRATED SCIENCE | PHOTOSYNTHESIS

Aim

Investigate whether light is needed for starch production in a green leaf.

Requirements

Green potted plant, opaque black paper, clips, forceps, beaker, hot water, ethanol in a boiling tube, electric hot-water bath, warm rinse water, white tile and iodine solution.

Procedure

1. Keep the plant in darkness for 48 hours so stored starch is used up.
2. Cover a middle strip of one leaf on BOTH surfaces with black paper and clips.
3. Place the plant in good light for several hours; keep it adequately watered.
4. Detach the leaf with forceps, remove the paper and note the covered strip.
5. Immerse the leaf in hot water to kill the tissue and stop chemical reactions.
6. Warm the leaf in ethanol in a hot-water bath to remove green chlorophyll.
7. Rinse the leaf in warm water to soften it, then spread it on a white tile.
8. Add iodine solution to the whole leaf and compare the two areas.

Ethanol is flammable: use indirect heating in a hot-water bath, never a naked flame.

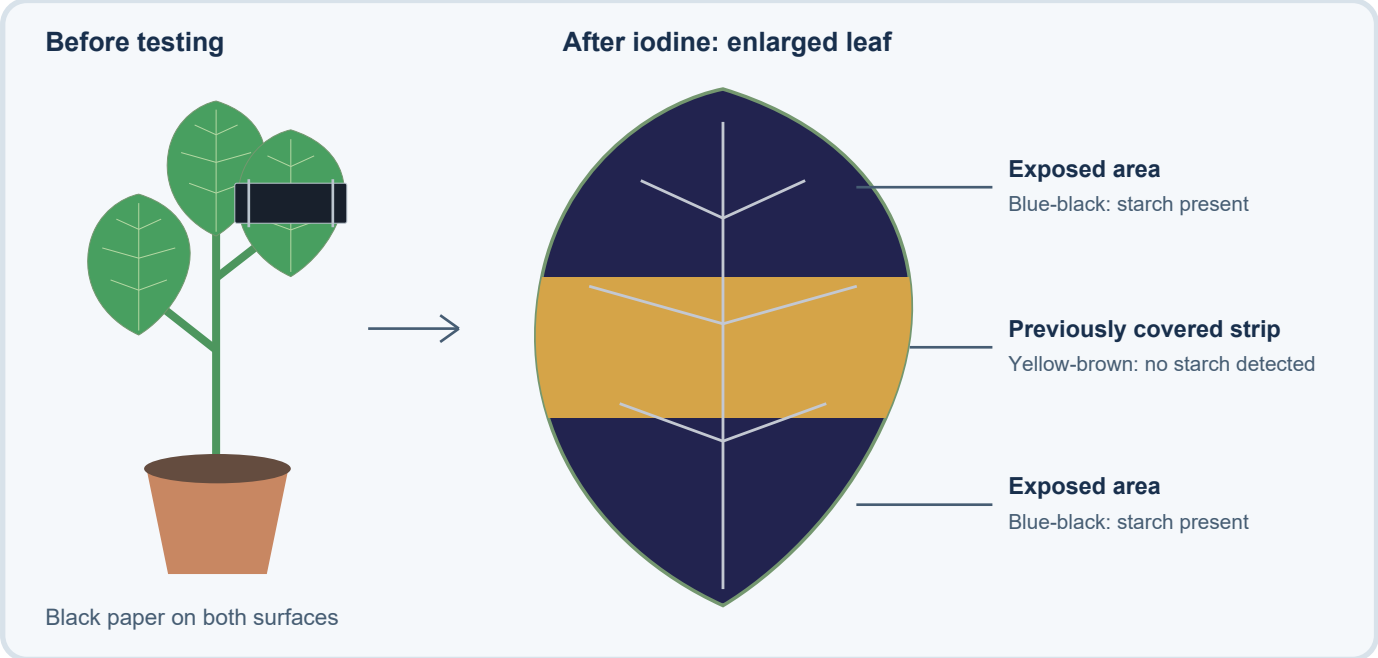


Figure 1. Starch-test colours in the exposed and covered areas of the same leaf.

Observation

The exposed areas turn blue-black after iodine is added, showing starch is present. The previously covered strip stays yellow-brown, showing no starch is detected there.

Conclusion

Light is needed for starch production in a green leaf. Both areas belonged to the same leaf, but only the areas exposed to light gave a positive starch test.